

ME594 3D Robotics
HW2 Answer to selected questions

1)

#H01

#rotate by 90 about x, rotate -90 about z, trans -1m about x

Ans

```
H01 = [[ 0.  1.  0.  0.]  
       [ 0.  0. -1.  0.]  
       [-1.  0.  0.  1.]  
       [ 0.  0.  0.  1.]]
```

#H02

#rotate by -90 about z, rotate by +90 about x, translate -1m about x

Ans

```
H02 = [[ 0.  0. -1.  0.]  
       [-1.  0.  0.  1.]  
       [ 0.  1.  0.  0.]  
       [ 0.  0.  0.  1.]]
```

#H12

#rotate by +90 about z, rotate by -90 about y, translate by -1 about y and -1 about x

Ans

```
H12 = [[ 0. -1.  0.  1.]  
       [ 0.  0. -1.  0.]  
       [ 1.  0.  0. -1.]  
       [ 0.  0.  0.  1.]]
```

2)

Ans: $\phi = \pi/4$; $\theta = \pi/6$; $\psi = \pi/3$;

Method:

```
import sympy as sp  
 $\psi, \theta, \phi = \text{sp.symbols('psi theta phi')}$   
Rz = sp.Matrix([  
    [sp.cos( $\psi$ ), -sp.sin( $\psi$ ), 0],  
    [sp.sin( $\psi$ ), sp.cos( $\psi$ ), 0],  
    [0, 0, 1]  
)  
and so on for Ry, Rz
```

$$R = R_z * R_y * R_x$$

```
sp.pprint(R)
```

In the expression for R (symbolic) you can solve for phi, theta, psi by comparing either Column 3 or Row 3 for the given R (numeric)